

IDRIS SADIQ

Senior Backend Engineer - Rust, AWS

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SUMMARY

Senior Backend Engineer with 10+ years delivering enterprise SaaS, workflow automation, data-intensive services, and cloud-integrated products, specializing in Rust 1.70-1.82 on AWS; combines API and data architecture, legacy modernization, secure delivery, production troubleshooting, observability, and technical leadership to improve reliability, performance, release confidence, and measurable business outcomes.

EXPERIENCE

SumatoSoft

Senior Software Engineer | January 2021 - December 2025

- Architected and delivered customer platforms with **Rust** on **AWS**, converting ambiguous operational requirements into technical designs, secure milestones, and supportable production releases.
- Modernized aging applications toward **Rust 1.70-1.82**, replacing tightly coupled modules with **domain-oriented services**, **asynchronous processing**, **idempotent workflows**, and **contract-first APIs** while preserving critical workflows through characterization, integration, and regression testing.
- Diagnosed difficult production incidents involving **connection-pool exhaustion**, **duplicate events**, **N+1 queries**, **deadlocks**, **retry storms**, and **idempotency failures**, correlating distributed traces, structured logs, browser profiles, queue metrics, and database execution plans to isolate root causes.
- Implemented new subscription, workflow, reporting, search, notification, and administrative capabilities across **REST** and **event-driven services**, **relational data stores**, **messaging**, **caching**, and **secure integrations**, balancing delivery speed with compatibility, accessibility, and operational readiness.
- Introduced **API versioning**, **schema validation**, **feature flags**, **idempotency controls**, and **automated rollback paths**, enabling gradual customer migrations without breaking older integrations or active sessions.
- Built secure delivery pipelines with **Docker**, **Kubernetes**, **Helm**, **Terraform**, **GitHub Actions**, **GitLab CI**, and **Argo CD**, standardizing environment promotion, evidence collection, and repeatable release verification.
- Strengthened identity and platform security with **OAuth 2.0**, **OpenID Connect**, **JWT validation**, **RBAC/ABAC**, **secrets management**, **rate limiting**, and **tamper-evident audit logging**.
- Mentored engineers through architecture reviews, pairing, code reviews, incident retrospectives, and practical testing standards, improving ownership while avoiding unnecessary process overhead.
- Improved API and user-flow performance by **39%** through query reshaping, targeted indexing, Redis caching, bundle reduction, batching, and removal of redundant integration calls during peak activity.

High Touch Technologies

Senior Software Engineer | June 2018 - December 2020

- Owned delivery of enterprise applications using **Rust** with period-appropriate cloud services, supporting operational workflows, customer portals, reporting, and third-party integrations across regulated environments.
- Refactored legacy modules into **domain-oriented services**, **asynchronous processing**, **idempotent workflows**, and **contract-first APIs**, reducing shared-state defects and allowing teams to release isolated functionality without rebuilding or redeploying unrelated application areas.
- Resolved production failures caused by **timeouts**, **lock contention**, **stale caches**, **malformed vendor payloads**, **partial outages**, and **configuration drift**, adding bounded retries and clearer failure contracts.
- Developed new case-management, scheduling, export, search, and role-based administration capabilities across **REST** and **event-driven services**, **relational data stores**, **messaging**, **caching**, and **secure integrations**, with backward-compatible data migrations and documented support procedures.
- Improved transaction correctness through **database constraints**, **narrower transaction scopes**, **idempotency keys**, **reconciliation jobs**, and **replay-safe queue consumers**, preventing duplicate or incomplete business actions.
- Expanded automated validation with **unit**, **integration**, **API contract**, **end-to-end**, **security**, and **performance tests**, catching edge cases that previously appeared only in customer-specific configurations.
- Partnered with product, QA, operations, and client teams during staged rollouts, using feature toggles, dashboards, and production checks to limit risk and communicate impact clearly.
- Reduced customer-reported workflow defects by **32%** after standardizing validation, observability, retry behavior, cache invalidation, and root-cause follow-up across recurring incident categories.

KEY ACHIEVEMENTS

Improved Product Performance

Improved API and user-flow performance by **39%** through query reshaping, targeted indexing, Redis caching, bundle reduction, batching, and removal of redundant integration calls during peak activity.

Reduced Production Defects

Reduced customer-reported workflow defects by **32%** after standardizing validation, observability, retry behavior, cache invalidation, and root-cause follow-up across recurring incident categories.

Accelerated Incident Resolution

Shortened average incident triage time by **28%** through structured logging, correlation identifiers, actionable dashboards, and runbooks that connected symptoms to likely service and data failure modes.

SKILLS

Backend: Rust 1.70-1.82, asynchronous processing, concurrency, dependency injection, validation, background jobs, secure service-to-service communication

Backend Testing & Reliability: pytest/JUnit/xUnit/Jest as applicable, Testcontainers, Pact, load testing, resilience patterns, circuit breakers, retries, backpressure, performance profiling

Cloud & Infrastructure: AWS, Docker, Kubernetes 1.24-1.31, Terraform 1.x, Linux, serverless services, networking, IAM, secrets, backup and disaster recovery

APIs & Architecture: REST, GraphQL, gRPC, WebSockets, event-driven architecture, microservices, modular monoliths, domain-driven design, CQRS, API versioning, idempotency

Data & Databases: PostgreSQL 13-17, MySQL 8, SQL Server, MongoDB 5-8, Redis 6-7, Elasticsearch/OpenSearch, schema design, indexing, query optimization, migrations

Quality & Delivery: GitHub Actions, Jenkins, GitLab CI, Azure DevOps, trunk-based development, feature flags, unit/integration/contract/E2E testing, code review, release management

Observability & Security: OpenTelemetry, Prometheus, Grafana, Datadog, CloudWatch, ELK/OpenSearch, Sentry, OAuth 2.0, OIDC, RBAC, SAST, SCA, threat modeling, incident response

Senior Engineering: architecture decision records, technical roadmaps, estimation, mentoring, cross-functional leadership, Agile delivery, production ownership, cost optimization, stakeholder communication

EXPERIENCE CONTINUED

Centric Tech Inc.

Software Engineer | September 2015 - May 2018

- Developed business applications and integration services with **Rust**, implementing stable interfaces, reusable modules, and period-appropriate deployment practices for internal and customer-facing workflows.
- Designed relational data models in PostgreSQL, MySQL, and SQL Server, adding selective indexes, query rewrites, and safer migration scripts for frequently used search and reporting operations.
- Investigated defects involving **incorrect totals, duplicate imports, null-handling errors, slow list queries, failed background jobs, and inconsistent date conversions** across production datasets.
- Implemented new dashboard, export, notification, account-management, and reconciliation features across **REST and event-driven services, relational data stores, messaging, caching, and secure integrations**, documenting edge cases and expected behavior for support and QA teams.
- Hardened external integrations with explicit timeouts, retry limits, token refresh handling, defensive parsing, and traceable audit records so vendor instability did not exhaust worker capacity.
- Automated build, test, and deployment checks with Jenkins, Docker, Bash, and environment scripts, improving repeatability across development, staging, and scheduled production releases.
- Shortened average incident triage time by **28%** through structured logging, correlation identifiers, actionable dashboards, and runbooks that connected symptoms to likely service and data failure modes.

Microsoft

Software Engineer Intern | September 2014 - August 2015

- Delivered carefully scoped service and internal-tool improvements using period-appropriate **C#, .NET, JavaScript, SQL Server, and Azure** technologies, emphasizing compatibility and maintainable implementation.
- Built utilities that automated repetitive operational and diagnostic steps, reducing manual error while producing consistent outputs that senior engineers could validate across environments.
- Improved troubleshooting by adding structured logs, contextual metadata, and targeted diagnostics around API and data-processing paths that previously produced ambiguous failure messages.
- Assisted with API integration changes by matching established request and response contracts, documenting edge cases, and preserving backward compatibility for dependent internal applications.
- Added focused unit and integration tests for bug fixes and new modules, incorporating code-review feedback on error handling, maintainability, security boundaries, and production readiness.
- Collaborated with engineers and product partners to translate requirements into testable acceptance criteria, reproduce reported defects, validate fixes, and document lightweight support runbooks.

SELECTED PROJECTS

Secure Transaction Services - Rust

Created versioned APIs and asynchronous workflows with **Rust 1.70-1.82**, relational data, Redis, messaging, idempotency, and OpenTelemetry; incorporated authorization, audit trails, contract tests, load tests, and rollback-safe database migrations for business-critical processing.

Cloud-Native Service Modernization - AWS

Decomposed a legacy service into bounded modules deployed on **AWS** with containers, infrastructure as code, managed data services, progressive delivery, and SLO-based monitoring; addressed connection pressure, slow queries, duplicate events, and release failures through measurable reliability work.

PROFESSIONAL DEVELOPMENT

Certification details were not included in the supplied profile; list only credentials personally earned and currently valid. Role-aligned professional development includes **AWS solution architecture, Kubernetes application delivery, secure software development, API design, and role-specific framework training**, supported by hands-on architecture, implementation, production support, mentoring, and security-focused engineering work.

EDUCATION

Harvard University | Bachelor's Degree, Computer Science | 2010 - 2014